Uniform permanence of endotactic reaction networks with measurable rates

October 8, 2026

Abstract

Let a finite reaction network be endotactic in the sense of Craciun, Nazarov and Pantea, and let its mass-action rate coefficients be arbitrary measurable functions of time with values almost everywhere in a fixed box $\prod_e [\underline{k}_e, \overline{k}_e]$ with $\underline{k}_e > 0$. Every positive stoichiometric compatibility class contains a nonempty compact convex set K, depending only on the network, the box and the class, with the following properties: every positive initial state has a unique global positive Carathéodory solution under every such rate history; K is invariant under every history; and for every compact set Q of positive initial states there is one deadline T_Q, chosen before the initial state and the history, after which every solution from Q stays in K. This is the classwise uniform permanence for ordinary endotactic networks with arbitrary signed real complexes and almost-everywhere bounded measurable histories. In particular it covers the nonnegative-complex domain of the Extended Permanence Conjecture of Craciun, Nazarov and Pantea (Conjecture 1.3 of Johnston, Pantea and Donnell). The proof combines a graph-independent finite affine barrier construction, imported from recent work of OpenAI on weakly reversible networks, with a new drift estimate for endotactic networks: at every active barrier normal, the reaction with the lowest adverse source is dominated by a single positive witness supplied by endotacticity, and the retained projection lengths give a margin uniform over the rate box. Trapping in convex plateaux and a finite iteration across logarithmic scales then yield the absorbing set and the uniform deadline. The real-complex theorem, including global positive Carathéodory solutions, uniqueness among positive solutions, invariance, compact-initial uniformity, null-set rate repair and initial-data continuous dependence, is established by strict Lean 4 checks with the standard axioms only. The positive periodic-solution and equilibrium corollaries are proved conventionally. A three-species network that is neither weakly reversible nor strongly endotactic illustrates the kind of explicit certificate the hypotheses allow: for six independently varying measurable coefficients in $[1/2, 2]$, every solution from $[0.1, 10]^3$ lies in $[0.4, 2.5]^3$ for all times $t \geq 347/16 < 22$, a staged bound whose clocked comparison steps and exact rational roundings are also machine-checked.

1 Introduction

A mass-action system assigns to each reaction $y \to y'$ of a finite network a positive rate coefficient and the monomial flux $k\, x^y$; the concentration vector then obeys $\dot{x} = \sum k_{y \to y'} x^y (y' - y)$ [12, 9]. Two long-standing questions concern the long-time behaviour of positive solutions. *Persistence* asks that no concentration approach zero along a trajectory; *permanence* asks for one compact set in the positive stoichiometric compatibility class that every solution in that class eventually enters and never leaves [2, 7]. The Global Attractor Conjecture asserts convergence of every positive trajectory of a complex-balanced system to the unique positive equilibrium in its stoichiometric compatibility class. It was proved for a single linkage class by Anderson [1] and in general by Craciun [5]; see

also [16, 11]. For weakly reversible networks with a single linkage class, permanence was proved by Boros and Hofbauer [3].

Craciun, Nazarov and Pantea [7] observed that the property that drives these results is not the graph condition of weak reversibility but a directional one. A network is *endotactic* if, in every direction $r \in \mathbb{R}^d$, every reaction whose vector points backwards ($r \cdot \nu_e < 0$) is accompanied by a reaction pointing forwards ($r \cdot \nu_j > 0$) whose source lies strictly lower in that direction ($r \cdot y_j < r \cdot y_e$). Weakly reversible networks are endotactic, but the converse fails. They proved that two-species endotactic systems are permanent even when the rate coefficients vary in time within a fixed positive box, and proposed the general statement as the Extended Permanence Conjecture; Johnston, Pantea and Donnell state it as Conjecture 1.3 of [13], whose Definition 3.1 is the directional witness condition used here. Gopalkrishnan, Miller and Shiu [11] proved permanence for *strongly* endotactic networks, where the forward witness must moreover start at a globally lowest source; Craciun and Deshpande [6] showed that endotactic networks are exactly those whose dynamics embed in toric differential inclusions. Ordinary endotacticity permits neutral reactions at globally extremal source levels and only asks for a witness below the adverse source; this is the hypothesis of the present paper.

Main result. Theorem 2.2 proves classwise uniform permanence for every finite ordinary endotactic network, every positive stoichiometric class, and every measurable rate history with values in a fixed positive compact box. The absorbing set K depends only on the network, the box and the class; the deadline for a compact set of initial states depends only on that set. The bounded measurable history satisfies the Carathéodory hypotheses on every compact positive state region. The proof constructs global absolutely continuous solutions satisfying the equation almost everywhere.

Method and attribution. The geometric input is a finite family of affine functions $x \mapsto r_\ell \cdot x + b_\ell(h)$ on the logarithmic box $[h, h^{-1}]^d$ whose lower envelope has the property that, at every point where a branch is active, the slope r_ℓ approximates the exponent vector $\log x / \log h$ to within a prescribed tolerance. This construction, with arbitrary pointwise positive (possibly discontinuous) tolerance, is Lemma 2.1 and Corollary 2.2 of OpenAI's preprint [14], reproved in full in Section 2 of the companion preprint [15]; both are public, with Lean sources, in the repository cited there. The second preprint uses it to prove permanence of weakly reversible networks with constant rates: the envelope is nondecreasing along trajectories, its ceiling plateau is a compact convex trapping set, and a scale-to-scale iteration gives a class-fixed absorbing set. Weak reversibility enters that argument only through a return-path flux estimate across level cuts.

The contribution of this paper is to replace that estimate by one that uses exactly the endotactic witness. At an active normal r, choose the adverse reaction with the lowest source; endotacticity supplies one forward reaction whose source lies below *every* adverse source, and source separation makes its monomial dominate the weighted sum of all adverse monomials at small scale. The weights are the actual projection lengths $|r \cdot \nu_e|$ and the box bounds, so the resulting drift margin is uniform over all rate vectors in the box, which is what measurable histories require (Section 4). A neutral normal, one that annihilates the stoichiometric subspace, has a constant branch on the class and is never active below the plateau. The remaining steps, strict trapping, global existence by continuation inside a compact positive plateau, and the dyadic scale iteration, follow the strategy of [15], adapted to absolutely continuous solutions and to a deadline uniform on compact initial sets (Section 5). The attribution is confined to the contributions and sources described here.

Consequences and example. Uniform continuous dependence on a compact invariant region gives, by Brouwer's theorem, a positive periodic solution under any almost-everywhere periodic admissible history and a positive equilibrium in every class of every autonomous endotactic system (Section 6). Section 7 treats a three-species network with six independently varying coefficients that is neither weakly reversible nor strongly endotactic. Its direct scalar structure yields an explicit certificate: every solution from the startup box $[0.1, 10]^3$ is inside the operating box $[0.4, 2.5]^3$ for all $t \geq 347/16 < 22$. The certificate is staged: the first coordinate recovers first, its improved floor accelerates the operational clock of the second coordinate, and that of the third. The general theorem supplies the qualitative guarantee; the example shows the quantitative constants that network structure can add.

Formal verification. Section 8 states exactly what has been machine-checked. The main theorem for arbitrary real complexes and a.e. admissible histories is verified end to end in Lean 4 with Mathlib [8, 17]. The solution interface is initial data, absolute continuity on every compact forward interval, and the equation almost everywhere. Global existence, positivity, uniqueness among positive solutions, invariance of K and the compact-initial uniform deadline are conclusions. The continuous-dependence estimate is checked in the same domain, and each root uses only the three standard axioms of Lean's classical logic. The fixed-point and feedback corollaries and the fixture-specific parts of the example remain conventional; the clocked comparisons and roundings have historical Lean checks.

2 Setting and main theorem

Let E be a finite index set of reactions. For $e \in E$ let $y_e, y'_e \in \mathbb{R}^d$ be its source and target complex and $\nu_e = y'_e - y_e$ its reaction vector. Parallel reactions, self-loops ($\nu_e = 0$) and the empty family are permitted. Put

$$S = \operatorname{span}_{\mathbb{R}}\{\nu_e : e \in E\}, \qquad P_c = (c + S) \cap \mathbb{R}^d_{>0} \quad (c \in \mathbb{R}^d_{>0}).$$

For $x \in \mathbb{R}^d_{>0}$ and $y \in \mathbb{R}^d$ write $x^y = \exp(\sum_i y_i \log x_i)$, the usual monomial when y has natural coordinates.

Definition 2.1 (Ordinary endotacticity)**.** The network is *endotactic* if for every $r \in \mathbb{R}^d$ and every $e \in E$,

$$r \cdot \nu_e < 0 \quad \Longrightarrow \quad \exists\, j \in E : \quad r \cdot \nu_j > 0 \ \text{ and } \ r \cdot y_j < r \cdot y_e. \tag{1}$$

All directions $r \in \mathbb{R}^d$ are tested, not only nonnegative ones (the latter would give the weaker "lower endotactic" condition of [16]). Nothing is required of neutral reactions ($r \cdot \nu_e = 0$), and the witness need not start at a globally lowest source; requiring that defines strong endotacticity [11, 13].

Fix numbers $0 < \underline{k}_e \leq \overline{k}_e < \infty$ and put

$$\mathcal{B} = \prod_{e \in E} [\underline{k}_e, \overline{k}_e], \qquad f_k(x) = \sum_{e \in E} k_e\, x^{y_e}\, \nu_e \qquad (k \in \mathcal{B},\ x \in \mathbb{R}^d_{>0}).$$

An *admissible history* is a function $k : [0, \infty) \to \mathbb{R}^E$ each of whose coordinates is Lebesgue measurable and which lies in $\mathcal{B}$ almost everywhere. A *solution* from $x_0 \in \mathbb{R}^d_{>0}$ under k is a function $x : [0, \infty) \to \mathbb{R}^d_{>0}$ that is absolutely continuous on every compact interval, satisfies $x(0) = x_0$, and satisfies $x'(t) = f_{k(t)}(x(t))$ for almost every t. Modifying k on a null set changes neither the class of admissible histories nor the solutions, so histories that are merely essentially bounded in $\mathcal{B}$ are covered after a modification on one common null set (E is finite).

Theorem 2.2 (Uniform endotactic permanence)**.** *Assume the network is endotactic. For every rate box $\mathcal{B}$ and every $c \in \mathbb{R}^d_{>0}$ there is a nonempty compact convex set $K \subset P_c$ such that:*

1. *every $x_0 \in P_c$ and every admissible history have a unique global solution, and it stays in P_c; uniqueness is among all solutions in the above sense, with no a priori boundedness assumption;*
2. *K is invariant: a solution starting in K remains in K for all $t \geq 0$, under every admissible history;*
3. *for every compact $Q \subset P_c$ there is $T_Q \geq 0$ such that every solution from every $x_0 \in Q$ under every admissible history lies in K for every $t \geq T_Q$.*

The set K is chosen before Q, and T_Q before x_0 and the history.

Compact containment in P_c gives one positive lower bound and one finite upper bound for every coordinate, valid after T_Q for all histories; these constants depend on the network, $\mathcal{B}$ and c, not on the history, and need not be uniform across classes or boxes. If $d = 0$ the state space is a point and there is nothing to prove; if $S = \{0\}$ (empty family or only self-loops) then $P_c = \{c\}$ and $K = \{c\}$ works with $T_Q = 0$. The proof for $d \geq 1$ occupies Sections 3–5. The argument handles a proper stoichiometric subspace without modification; the class enters through the neutral normals discussed in Section 4.

3 The finite affine geometric input

For $0 < h < 1$ put $B_h = [h, h^{-1}]^d$. Every $x \in B_h$ can be written uniquely as $x = h^p$ coordinatewise, with $p = \log x / \log h \in [-1, 1]^d$. Let $D \subset \mathbb{R}^d$ be the finite set consisting of the coordinate vectors e_i together with all nonzero ordered differences of sources $y_e - y_j$ $(e, j \in E)$; differences between sources in different connected components of the reaction graph are included. For $r \in [-1, 1]^d$ define the tolerance

$$\epsilon(r) = \min\Big(\{1/2\} \cup \Big\{\frac{|r \cdot v|}{2\,\|v\|_1} : v \in D,\ r \cdot v \neq 0\Big\}\Big). \tag{2}$$

Zero scalar products are omitted, so $\epsilon(r) > 0$ for every r and $\epsilon(0) = 1/2$; no continuity of ϵ is assumed or true.

Lemma 3.1 (Affine approximation with discontinuous tolerance)**.** *There are a nonempty finite label set I, slopes $r_\ell \in [-1, 1]^d$ $(\ell \in I)$ independent of h, offsets $b_\ell(h) \in \mathbb{R}$ defined for small $h > 0$, and $h_g \in (0, 1)$ such that, whenever $0 < h < h_g$, $p \in [-1, 1]^d$, and ℓ attains the minimum in*

$$F_h(h^p) = \min_{m \in I}\{r_m \cdot h^p + b_m(h)\},$$

one has $\|p - r_\ell\|_\infty < \epsilon(r_\ell)$.

This is the uniform form, Corollary 2.2, of Lemma 2.1 in [14] (reproved in [15, Section 2]) applied with $m = d$, $[a, b] = [-1, 1]$ and the tolerance (2). The lemma is graph independent: it concerns only the finite set D of comparison vectors. Its sequential statement covers every fixed label, including ties; the uniform statement follows because a failure sequence $h_\nu \to 0$, $p_\nu \in [-1, 1]^d$ with a violating active label would, by finiteness and compactness, have a subsequence with one fixed label and convergent exponents, contradicting the sequential bound for that label, the tolerance being evaluated at the fixed slope throughout. The family, hence its finite set of slopes, is fixed

before the threshold h_g. The Lean port of this construction is discussed in Section 8; it is a proved dependency of the main theorem, not an axiom.

Fix the family and define the positive separation

$$H = \min\big(\{1\} \cup \{|r_\ell \cdot v|/2 : \ \ell \in I,\ v \in D,\ r_\ell \cdot v \neq 0\}\big) > 0. \tag{3}$$

Let $x = h^p \in B_h$ with $0 < h < h_g$, let ℓ be active at x and write $r = r_\ell$. For $v \in D$ with $r \cdot v \neq 0$,

$$|(p - r) \cdot v| \leq \|p - r\|_\infty \, \|v\|_1 < \epsilon(r) \, \|v\|_1 \leq |r \cdot v|/2,$$

so $p \cdot v$ has the sign of $r \cdot v$ and $|p \cdot v| > |r \cdot v|/2 \geq H$. Since $x^y = h^{p \cdot y}$ and $0 < h < 1$, this gives the *source-ordering estimate*

$$r \cdot y_j < r \cdot y_e \quad \Longrightarrow \quad x^{y_e} \leq h^H x^{y_j} \qquad (x = h^p \in B_h,\ r = r_\ell \text{ active at } x), \tag{4}$$

and the *coordinate-sign estimate*: if $r_i \neq 0$ then p_i has the sign of r_i and $|p_i| > H$. The gap H is taken after the finite family is fixed; no separation uniform over a continuum of directions is asserted, and exact zero comparisons impose nothing.

4 Endotactic drift and the convex plateau

Call a slope r *neutral* if $r \cdot \nu_e = 0$ for every e, that is, $r \perp S$.

Lemma 4.1 (One witness below all adverse sources)**.** *Let r be nonneutral. There is $j \in E$ with $r \cdot \nu_j > 0$ and $r \cdot y_j < r \cdot y_e$ for every e with $r \cdot \nu_e < 0$.*

Proof. If adverse reactions exist, pick e_- minimizing $r \cdot y_e$ among them. Condition (1) gives j with $r \cdot \nu_j > 0$ and $r \cdot y_j < r \cdot y_{e_-} \leq r \cdot y_e$ for every adverse e. If no adverse reaction exists, nonneutrality gives some j with $r \cdot \nu_j > 0$, and there is no inequality to check. □

For each nonneutral slope $r = r_\ell$ of the fixed family choose such a witness $j = j(\ell)$ and set

$$\alpha_r = \underline{k}_j \, (r \cdot \nu_j) > 0, \qquad \beta_r = \sum_{e:\, r \cdot \nu_e < 0} \overline{k}_e \, |r \cdot \nu_e| \ \geq 0.$$

Let $k \in \mathcal{B}$, $0 < h < h_g$, $x = h^p \in B_h$, and let ℓ be active at x. Keeping the witness term, discarding the other nonnegative terms, bounding the coefficients by the box, and applying (4) to every adverse source gives

$$r \cdot f_k(x) \ \geq \ \alpha_r x^{y_j} - \sum_{r \cdot \nu_e < 0} \overline{k}_e |r \cdot \nu_e| \, x^{y_e} \ \geq \ x^{y_j} (\alpha_r - h^H \beta_r). \tag{5}$$

The projection lengths $|r \cdot \nu_e|$ are retained; replacing them by a count of adverse reactions would be wrong when projected lengths differ. Since the family is finite, there is $h_* \in (0, h_g)$ such that $h^H \beta_r \leq \alpha_r/2$ for all nonneutral slopes whenever $0 < h < h_*$. Let $D_* = \max(\{0\} \cup \{\sum_i |y_{e,i}| : e \in E\})$. For arbitrary signed sources, $x \in B_h$ implies $x^y \geq h^{D_*}$ for every source y. If nonneutral slopes exist let $\alpha_* > 0$ be the least of their α_r. Then for $0 < h < h_*$, every $k \in \mathcal{B}$, and every $x \in B_h$,

$$r_\ell \cdot f_k(x) \ \geq \ \delta_h := \tfrac{1}{2} \alpha_* h^{D_*} > 0 \qquad \text{for every nonneutral label } \ell \text{ active at } x. \tag{6}$$

A neutral slope has $r \cdot f_k(x) = 0$ identically. If every slope is neutral the strict statements below are vacuous and any $\delta_h > 0$ may be used. All choices so far depend on the network and $\mathcal{B}$ only.

The plateau at a class representative. Now fix $c \in \mathbb{R}^d_{>0}$. Decrease the threshold h_* further (depending on c) so that $\|\log c/\log h\|_\infty < H$ for all admissible h. If a label ℓ is active at c and $r_{\ell,i} \neq 0$ for some i, the coordinate-sign estimate would give $|\log c_i/\log h| > H$; hence every slope active at c is zero. Pick such a label and put $M_h = F_h(c)$. Its branch is the constant M_h, so $F_h \le M_h$ on all of $\mathbb{R}^d$, and every other constant branch is $\ge M_h$. Define the *plateau*

$$C_h = \{x \in \mathbb{R}^d : F_h(x) = M_h\} = \bigcap_{\ell \in I} \{x : r_\ell \cdot x + b_\ell(h) \ge M_h\}, \tag{7}$$

a closed convex polyhedron containing c.

Lemma 4.2 (Compact plateau and classwise strict drift)**.** *For every admissible h,*

$$C_h \subset \operatorname{int} B_{\sqrt{h}}, \qquad C_h \text{ is compact.}$$

If $x \in (P_c \cap B_h) \setminus C_h$, every label active at x is nonneutral, so $r_\ell \cdot f_k(x) \ge \delta_h$ for every active ℓ and every $k \in \mathcal{B}$.

Proof. Let $x = h^p \in C_h \cap B_h$. The chosen zero-slope branch equals $M_h = F_h(x)$, so it is active at x, and Lemma 3.1 with $\epsilon(0) = 1/2$ gives $|p_i| < 1/2$, that is $\sqrt{h} < x_i < h^{-1/2}$: $C_h \cap B_h \subset \operatorname{int} B_{\sqrt{h}}$. Suppose some $z \in C_h$ lay outside B_h. Since $c \in \operatorname{int} B_h$ and C_h is convex, the segment from c to z lies in C_h; its first point on the boundary of B_h would belong to $C_h \cap B_h$ yet not to $\operatorname{int} B_{\sqrt{h}}$, a contradiction. Hence $C_h \subset \operatorname{int} B_{\sqrt{h}}$; it is closed and bounded, hence compact.

If r_ℓ is neutral then $r_\ell \cdot x = r_\ell \cdot c$ on $c + S$, so the branch ℓ is constant on the class with value $\ge F_h(c) = M_h$. At $x \in P_c$ with $F_h(x) < M_h$ this branch is not minimal, so ℓ is not active. The remaining active labels are nonneutral and (6) applies. □

Neutrality of individual reactions is not the issue here; what matters is whether a slope annihilates the whole subspace S. A nonneutral slope may well have many neutral reactions, and the witness lemma handles them.

5 Solutions, trapping, and uniform absorption

5.1 Carathéodory solutions for measurable histories

On any compact positive box the finitely many monomials x^{y_e} and their state derivatives $\partial_i x^y = y_i x^y / x_i$ are bounded. Hence $(t, x) \mapsto f_{k(t)}(x)$ is measurable in t, smooth in x on $\mathbb{R}^d_{>0}$, and bounded and Lipschitz in x on each compact positive box uniformly over all histories with values in $\mathcal{B}$. These are the standard Carathéodory conditions [4, Chapter 2], [10, §1]; the following records the argument in the form used by the formal proof. Fix $x_0 \in \mathbb{R}^d_{>0}$, $s \ge 0$, a closed ball of radius ρ about x_0 inside $\mathbb{R}^d_{>0}$, a bound A for $\|f_k\|$ and a Lipschitz constant L for f_k on that ball, uniform in $k \in \mathcal{B}$. For $\eta > 0$ with $A\eta \le \rho$ and $L\eta < 1$, the Picard map

$$(\mathcal{T}u)(t) = x_0 + \int_s^t f_{k(v)}(u(v))\, dv \qquad (t \in [s, s+\eta])$$

sends continuous paths in the ball to continuous paths in the ball and is a contraction; its fixed point is an integral solution, hence absolutely continuous and a solution of the equation almost everywhere, and any two solutions agree on $[s, s+\eta]$ by the same estimate. Repeating the construction yields a unique maximal positive solution, and all its integral increments lie in the closed subspace S, so it stays in the affine class $c + S$ as long as it exists.

Let x be a solution segment in B_h and put $g(t) = F_h(x(t))$. The minimum of finitely many affine functions is Lipschitz with constant at most d in the maximum norm, so g is absolutely continuous on every compact interval. At almost every t the curve, g, and each branch composition $t \mapsto r_\ell \cdot x(t) + b_\ell(h)$ are differentiable and the equation holds. If ℓ is active at such a t, the nonnegative differentiable function (branch minus envelope) vanishes at t, so its derivative there is zero and $g'(t) = r_\ell \cdot f_{k(t)}(x(t))$. This handles ties among active branches without selecting one. Hence, while $x(t) \in P_c \cap B_h$,

$$g'(t) \geq 0 \quad \text{a.e.}, \qquad g'(t) \geq \delta_h \quad \text{a.e. on } \{t : g(t) < M_h\}, \tag{8}$$

by Lemma 4.2 and the neutral-slope identity. These are almost-everywhere statements; nothing is assumed at switching times of the active set.

5.2 Trapping and global existence

Let $x_0 \in C_h \cap P_c$. Then $x_0 \in \operatorname{int} B_h$. While the maximal solution stays in B_h, (8) and the fundamental theorem for absolutely continuous functions make g nondecreasing; since $g(0) = M_h$ and $g \leq M_h$, the solution stays in C_h. A first exit from $\operatorname{int} B_h$ would be a point of C_h on the boundary of B_h, contradicting Lemma 4.2. So the maximal solution stays in the compact positive set $C_h \cap (c + S)$.

If its maximal interval of existence ended at a finite b, the field bound A on a positive box containing C_h would give $\|x(t) - x(s)\| \leq A|t - s|$, so $x(t)$ would converge as $t \uparrow b$ to a point x_b of that compact set; local existence from x_b under the shifted history and concatenation of the integral equations would extend the solution beyond b. Hence the solution is global and positive. The same argument shows that $C_h \cap P_c$ is invariant for every admissible history, and uniqueness among all solutions follows from local Lipschitz uniqueness on overlapping compact intervals. For natural-number complexes the field is polynomial on all of $\mathbb{R}^d$, and the formal uniqueness statement makes no positivity or boundedness assumption on the competing solution.

Finite entry. Suppose a global positive solution lies in $P_c \cap B_h$ for all $t \geq a$, and let $A_h \leq F_h(x(a))$. If $x(t) \notin C_h$ for all $t \in [a, a + (M_h - A_h)/\delta_h + 1]$, integrating the strict part of (8) would raise g by more than $M_h - A_h$, contradicting $g \leq M_h$. So the solution enters C_h no later than

$$a + \frac{M_h - A_h}{\delta_h} + 1, \tag{9}$$

and by invariance it stays there. The added 1 is a convenient strict slack, not a numerical approximation.

5.3 The dyadic scale iteration

Choose one $p \in (0, 1)$ below all thresholds above; it depends only on the network, $\mathcal{B}$ and c. Define

$$h_n = p^{2^n}, \qquad C_n = C_{h_n}, \qquad F_n = F_{h_n}, \qquad M_n = F_n(c) \qquad (n \geq 0).$$

These scales decrease to 0 and satisfy $\sqrt{h_{n+1}} = h_n$, so Lemma 4.2 gives

$$C_{n+1} \subset \operatorname{int} B_{h_n}. \tag{10}$$

The plateaux need not be nested; only (10) is used. By compactness $A_n = \min_{x \in C_{n+1}} F_n(x)$ exists, and $A_n \le M_n$ because $c \in C_{n+1}$. A solution trapped in $C_{n+1} \cap P_c$ from time a on stays in B_{h_n}, so by (9) it enters, and then stays in, C_n by time $a + D_n$ where

$$D_n = \frac{M_n - A_n}{\delta_{h_n}} + 1, \qquad T_N = \sum_{n=0}^{N-1} D_n. \tag{11}$$

By induction, a solution trapped in $C_N \cap P_c$ from time 0 lies in C_0 for all $t \ge T_N$, under every admissible history.

Proof of Theorem 2.2. Let $x_0 \in P_c$. The numbers $|\log x_{0,i}|$ are finite and $|\log h_N| \to \infty$, so there is N with $\|\log x_0 / \log h_N\|_\infty < H$; by the coordinate-sign estimate every slope active at x_0 at scale h_N is zero, so $F_N(x_0)$ equals a constant branch, which is $\ge M_N$, while $F_N \le M_N$; thus $x_0 \in C_N$. Trapping gives the unique global positive solution in $C_N \cap P_c$, and the iteration places it in C_0 by time T_N. Put

$$K = C_0 \cap (c + S).$$

It contains c, is compact and convex, and lies in $\mathbb{R}^d_{>0}$ by Lemma 4.2, so $K \subset P_c$; it is invariant because $C_0 \cap P_c$ and $c + S$ are. This proves (1) and (2) and the absorption of each individual initial state.

For (3) let $Q \subset P_c$ be compact. The coordinate logarithms are continuous on Q, so $|\log x_i| \le A_Q$ on Q for some $A_Q \ge 0$, and one index N with $A_Q / |\log h_N| < H$ places all of Q in C_N. The deadline $T_Q = T_N$ then works for every $x_0 \in Q$ and every admissible history. This common starting plateau is what gives uniformity; the pointwise existence of entry times together with compactness of Q would not. □

Formula (11) is a rigorous bound in terms of the barrier data (M_n, A_n, δ_{h_n}). The affine construction does not make these data efficiently computable, and Section 7 shows that direct scalar structure, when available, yields far better constants.

5.4 Signed real complexes and null-set rate repair

The theorem allows $y_e, y'_e \in \mathbb{R}^d$. The identity $x^y = \exp(\sum_i y_i \log x_i)$ gives smoothness and bounded derivatives on every compact positive box, including for negative exponents. On B_h with $0 < h < 1$, $|\log x_i| \le -\log h$, so $x^y \ge h^{\sum_i |y_i|} \ge h^{D_*}$. The comparison set remains finite and Lemma 3.1 concerns only $p \in [-1, 1]^d$. No nonnegativity of complexes is used in the drift or trapping arguments. Uniqueness is asserted among positive solutions: on each finite time interval the two continuous positive curves admit a common strictly positive compact box, on which local Lipschitz uniqueness applies. Boundary regularity is not required. Natural complexes recover the usual polynomial field, and the preserved historical theorem also gives uniqueness among ambient solutions.

For a.e. admissible rates, choose a measurable representative of each coordinate and clamp it to $[\underline{k}_e, \overline{k}_e]$. Finiteness of E gives one common exceptional null set. The resulting history is measurable and bounded pointwise and agrees with the original history off that set. The two vector fields therefore agree there for every state simultaneously. This transfers the AC/a.e.-ODE predicate for every candidate curve, hence existence, positive-solution uniqueness and invariance. The absorbing set and the deadline were chosen uniformly for the whole rate box before selecting a history, so they are independent of this repair. Both extensions are included in the fresh formal roots described in Section 8.

6 Immediate consequences

Proposition 6.1 (Uniform continuous dependence)**.** *Let $C \subset \mathbb{R}^d_{>0}$ be compact. There is $L \geq 0$ depending on the network, $\mathcal{B}$ and C only, such that any two solutions x, y under the same admissible history that both remain in C satisfy*

$$\|x(t) - y(t)\| \leq \|x(0) - y(0)\| \, e^{Lt} \qquad (t \geq 0). \tag{12}$$

In particular this applies on the invariant set K of Theorem 2.2.

Proof. Let L be a Lipschitz constant of f_k on C, uniform over $k \in \mathcal{B}$. The two integral equations give $q(t) \leq q(0) + L \int_0^t q(s)\, ds$ for the continuous function $q(t) = \|x(t) - y(t)\|$. Put $U(t) = q(0) + L \int_0^t q(s)\, ds$; then $q \leq U$, U is C^1 with $U' = Lq \leq LU$, and $(e^{-Lt}U)' \leq 0$ gives $q \leq U \leq q(0)e^{Lt}$. Only the auxiliary primitive U is differentiated pointwise; the measurable field enters in integral form. □

Corollary 6.2 (Periodic operation)**.** *Let $\tau > 0$ and let an admissible history satisfy $k(t+\tau) = k(t)$ for almost every $t \geq 0$. Then there is a positive solution with $x(t+\tau) = x(t)$ for all $t \geq 0$, taking all its values in K.*

Proof. The time-τ map $\Phi_\tau(z) = x_z(\tau)$ sends K to itself by invariance and is continuous by Proposition 6.1. Brouwer's fixed-point theorem, applied in the affine hull of the nonempty compact convex set K, gives $z \in K$ with $\Phi_\tau(z) = z$. The shifted curve $t \mapsto x_z(t+\tau)$ is absolutely continuous on compact intervals, starts at z, and satisfies the equation almost everywhere because translation preserves null sets and the coefficients agree almost everywhere; by uniqueness it equals x_z. □

This is an existence statement; uniqueness, stability and entrainment of the periodic regime are not claimed.

Corollary 6.3 (Autonomous equilibria)**.** *For every constant $k \in \mathcal{B}$, every positive stoichiometric class of an endotactic network contains a positive equilibrium of f_k.*

Proof. Write $f = f_k$ and Φ_s for its flow. For $\tau_n = 1/(n+1)$, Brouwer gives $z_n \in K$ with $\Phi_{\tau_n}(z_n) = z_n$; pass to a subsequence $z_n \to z_* \in K$. With A a bound for $\|f\|$ on K, $\|\Phi_s(z_n) - z_n\| \leq As$ for $0 \leq s \leq \tau_n$, so these short arcs converge uniformly to z_*, and uniform continuity of f on K gives

$$0 = \frac{\Phi_{\tau_n}(z_n) - z_n}{\tau_n} = \frac{1}{\tau_n} \int_0^{\tau_n} f(\Phi_s(z_n))\, ds \longrightarrow f(z_*).$$

□

A fixed point of a single time-τ map could lie on a periodic orbit; the shrinking times are needed. The permanence-plus-Brouwer route to equilibria has a two-species antecedent in [7, Remark 6.2]; here it is available at the general endotactic scope.

Corollary 6.4 (Bounded multiplicative modulation)**.** *If $k_e(t) = \bar{k}_e \theta_e(t)$ with $\bar{k}_e > 0$ and measurable $0 < \underline{\theta}_e \leq \theta_e(t) \leq \overline{\theta}_e < \infty$, then Theorem 2.2 applies with $\underline{k}_e = \bar{k}_e \underline{\theta}_e$ and $\overline{k}_e = \bar{k}_e \overline{\theta}_e$. No slow-variation, switching-rate or independence assumption is involved.*

Proposition 6.5 (State feedback with local hypotheses)**.** *Let $k : [0, \infty) \times \mathbb{R}^d_{>0} \to \mathcal{B}$ be jointly measurable and such that the closed-loop field $(t, x) \mapsto f_{k(t,x)}(x)$ is bounded and Lipschitz in x on every compact positive box, uniformly in t. Then every positive initial state has a unique global positive closed-loop solution, K is invariant for it, and the deadline T_Q of Theorem 2.2 applies to closed-loop solutions from Q.*

Proof. The closed-loop field satisfies the Carathéodory conditions, so a local solution exists and is unique. Along it, the evaluated history $t \mapsto k(t, x(t))$ is measurable with values in $\mathcal{B}$, so the trapping argument of Section 5.2 applies to the local solution and confines it to a compact positive plateau; the field bound then gives a limit at any finite maximal time and local existence continues the solution. Only after global existence is established is the evaluated history a global admissible history, to which Theorem 2.2 applies; its bounds and the starting plateau do not depend on the feedback law. □

7 A quantitative three-species example

Consider the nondimensional system

$$\dot{a} = c\,(k_1 - k_2 a^2), \qquad \dot{b} = ab\,(k_3 - k_4 b^2), \qquad \dot{c} = bc\,(k_5 - k_6 c^2), \qquad \tfrac{1}{2} \le k_i(t) \le 2, \tag{13}$$

with six independent measurable coefficients. It is the mass-action system of the six reactions of Figure 1.

$$\begin{array}{ccc} C \xrightarrow{k_1} A + C & A + B \xrightarrow{k_3} A + 2B & B + C \xrightarrow{k_5} B + 2C \\ 2A + C \xrightarrow{k_2} A + C & A + 3B \xrightarrow{k_4} A + 2B & B + 3C \xrightarrow{k_6} B + 2C \end{array}$$

Figure 1: Three inward reaction pairs with terminal targets. Each column has sources u and $u+2v$ and common target $u + v$, with $v = e_A, e_B, e_C$ respectively. For the direction $r = (0, 1, 0)$ the globally lowest sources are the neutral ones in the first column.

7.1 Structural classification

For a pair $u \to u + v$, $u + 2v \to u + v$ and a direction r: if $r \cdot v > 0$ the first reaction is a forward witness with source u below the adverse source $u + 2v$; if $r \cdot v < 0$ the roles are reversed; if $r \cdot v = 0$ neither reaction is adverse. Hence each pair, and their union, is endotactic in every real direction. The reaction vectors are $\pm e_A, \pm e_B, \pm e_C$, so $S = \mathbb{R}^3$ and the positive class is the whole open orthant. At $r = (0, 1, 0)$ the globally lowest sources C and $2A + C$ are neutral while the forward witness $A + B$ starts at level 1, so the network is not strongly endotactic. Each target is terminal, so the network is not weakly reversible. It has nine complexes, three linkage classes, rank 3 and deficiency $9 - 3 - 3 = 3$; the displayed reaction graph is acyclic and cannot be complex balanced at positive rates, since a terminal target receives positive influx and has no outflux (this concerns the displayed realization; dynamically equivalent realizations are not excluded here). Full rank excludes any nonzero linear conservation law.

7.2 Invariant boxes

For a positive initial state put

$$m = \min(a_0, b_0, c_0, 1/2), \qquad M = \max(a_0, b_0, c_0, 2). \tag{14}$$

The box $[m, M]^3$ is invariant for every admissible history: on a lower face of a coordinate the bracket in its equation is at least $1/2 - 2m^2 \ge 0$, on an upper face at most $2 - M^2/2 \le 0$, and the remaining factors are positive. To make this a rigorous almost-everywhere argument even with weak face signs, clamp the state to the box, construct the bounded Lipschitz integral solution of

the clamped field, and note that the positive part of the distance of each coordinate to its interval is absolutely continuous with nonpositive derivative a.e. where positive and zero derivative a.e. where zero; starting from zero it stays zero, so the clamped solution solves (13) and uniqueness proves invariance. The same face signs, now strict ($9/50 > 0$ and $-9/8 < 0$), make the operating box $[2/5, 5/2]^3$ invariant. Enlarging the operating box beyond $[1/2, 2]^3$ matters: with $k_1 = 1/2$, $k_2 = 2$ and $a_0 < 1/2$ the first coordinate approaches $1/2$ from below without reaching it in finite time.

7.3 Clocks and scalar comparisons

Assume $(a_0, b_0, c_0) \in [1/10, 10]^3$; by invariance all three coordinates stay in $[1/10, 10]$. Each coordinate has an *operational clock*

$$\sigma_a(t) = \int_0^t c(s)\,ds, \qquad \sigma_b(t) = \int_0^t a(s)\,ds, \qquad \sigma_c(t) = \int_0^t b(s)\,ds,$$

an absolutely continuous nondecreasing function whose derivative is the prefactor of the corresponding equation. The startup box bounds every clock speed below by $1/10$: $\sigma(t) \geq t/10$ for each clock.

The first coordinate (conventional). Since σ_a is locally bi-Lipschitz with $\sigma_a' \geq 1/10$, the substitution $t \mapsto \sigma_a$ is valid for absolutely continuous equations and preserves the almost-everywhere statements. In the clock $\sigma = \sigma_a$,

$$\tfrac{1}{2} - 2a^2 \;\leq\; \frac{da}{d\sigma} \;\leq\; 2 - \tfrac{1}{2}a^2 \qquad \text{a.e.}$$

The Riccati comparison equations $a' = \frac{1}{2} - 2a^2$ from $a(0) = 1/10$ and $a' = 2 - \frac{1}{2}a^2$ from $a(0) = 10$ have the solutions $\frac{1}{2}\tanh(\sigma + \operatorname{atanh}\frac{1}{5})$ and $2\coth(\sigma + \operatorname{atanh}\frac{1}{5})$, and the positive-part argument with a local Lipschitz bound (valid for measurable coefficients) gives

$$\tfrac{1}{2}\tanh\bigl(\sigma + \operatorname{atanh}\tfrac{1}{5}\bigr) \leq a \leq 2\coth\bigl(\sigma + \operatorname{atanh}\tfrac{1}{5}\bigr). \tag{15}$$

Both $a \geq 2/5$ and $a \leq 5/2$ hold once $\sigma \geq \operatorname{atanh}\frac{4}{5} - \operatorname{atanh}\frac{1}{5} = \frac{1}{2}\log 6$, hence for all

$$t \;\geq\; T_a := 5\log 6 \;<\; 9, \tag{16}$$

and they persist because the comparison envelopes move inward. The rounding $5\log 6 < 9$ is the exact inequality $e^{9/5} > \sum_{j=0}^{6}(9/5)^j/j! = 7542629/1250000 > 6$.

The second and third coordinates (clocked linear comparison). Let $u \in \{b, c\}$ with clock $\sigma \in \{\sigma_b, \sigma_c\}$ and coefficients $(k_+, k_-) \in \{(k_3, k_4), (k_5, k_6)\}$, so that $u' = \sigma' u(k_+ - k_- u^2)$ almost everywhere. Because u is confined to $[1/10, 10]$ and $v \mapsto v^{-2}$ is Lipschitz there, $z = u^{-2}$ is absolutely continuous, and by the chain rule, for almost every t,

$$z' = -2u^{-3}u' = \sigma'(-2k_+ z + 2k_-), \qquad \text{so} \qquad \sigma'(1 - 4z) \;\leq\; z' \;\leq\; \sigma'(4 - z), \tag{17}$$

using $k_+ \geq 1/2$, $k_- \leq 2$, $z > 0$ and $\sigma' \geq 0$. The relevant comparison is stated in the clock directly, without changing the time variable.

Lemma 7.1 (Clocked linear comparison)**.** *Let $z, \sigma : [0,T] \to \mathbb{R}$ be absolutely continuous, $\beta \in \mathbb{R}$, and suppose $z'(t) \le \sigma'(t)\big(\beta - z(t)\big)$ for almost every $t \in [0,T]$. Then*

$$z(T) \;\le\; \beta + (z(0) - \beta)e^{-(\sigma(T)-\sigma(0))}.$$

The symmetric statement with $\ge$ holds for a lower bound.

Proof. σ is continuous, hence bounded by some M on $[0,T]$, and exp is Lipschitz on $(-\infty, M]$; so $w = e^{\sigma}(z-\beta)$ is absolutely continuous as a product of absolutely continuous functions. Almost everywhere $w' = e^{\sigma}\big(\sigma'(z-\beta) + z'\big) \le 0$. Integrating, $w(T) \le w(0)$, which is the claim after division by $e^{\sigma(T)} > 0$. □

This lemma, including the composition step and the integration, is verified in Lean (`ac_clocked_linear_upper`; the constant-speed case is `ac_linear_upper`). Applied to (17) with $z(0) = u_0^{-2} \in [1/100, 100]$:

$$\tfrac{1}{4} - \tfrac{6}{25}e^{-4\sigma(t)} \;\le\; z(t) \;\le\; 4 + 96\,e^{-\sigma(t)} \qquad (t \ge 0). \tag{18}$$

The upper bound on z is a *lower* bound on u: $u \ge 2/5$ as soon as $\sigma(t) \ge \log(128/3)$, and persistently so because σ is nondecreasing. The lower bound gives $u \le 5/2$ as soon as $\sigma(t) \ge \frac{1}{4}\log(8/3)$.

7.4 Two certified deadlines

With the uniform clock floor $\sigma(t) \ge t/10$ alone, (18) gives $u \ge 2/5$ for $t \ge T_* := 10\log(128/3)$ and $u \le 5/2$ for $t \ge \frac{5}{2}\log(8/3)$; together with (16) this is the one-stage certificate.

Proposition 7.2 (One-stage certificate)**.** *Every solution of (13) from $[1/10, 10]^3$ lies in $[2/5, 5/2]^3$ for every $t \ge T_* = 10\log(128/3)$, and $T_* < 38$.*

Proof. All deadlines in the preceding paragraph are at most T_* since $5\log 6 < T_*$ and $\frac{5}{2}\log(8/3) < T_*$. The rounding is the exact rational inequality

$$e^{19/5} \;\ge\; \sum_{j=0}^{9} \frac{(19/5)^j}{j!} = \frac{1968639501739}{44296875000} \;>\; \frac{128}{3}, \tag{19}$$

so $\log(128/3) < 19/5$ and $T_* < 38$. No floating-point value enters. □

The one-stage bound of Proposition 7.2 wastes the fact that the first coordinate recovers early. Once $a \ge 2/5$, the clock σ_b runs at speed at least $2/5$, and once $b \ge 2/5$, so does σ_c. Feeding these floors back in sequence gives a staged certificate; the order matters, since assuming all three floors from the start would be circular.

Theorem 7.3 (Staged certificate)**.** *Every solution of (13) from $[1/10, 10]^3$ under every admissible history satisfies*

$$a(t) \in [\tfrac{2}{5}, \tfrac{5}{2}] \;\; (t \ge 9), \qquad b(t) \in [\tfrac{2}{5}, \tfrac{5}{2}] \;\; (t \ge \tfrac{65}{4}), \qquad c(t) \in [\tfrac{2}{5}, \tfrac{5}{2}] \;\; (t \ge \tfrac{347}{16}).$$

In particular every such solution lies in $[0.4, 2.5]^3$ for all $t \ge 347/16 = 21.6875$, hence for all $t \ge 22$.

Proof. Upper bounds. By (16), $a \le 5/2$ for $t \ge 9$. For $u \in \{b, c\}$ the lower half of (18) with $\sigma \ge t/10$ gives $z \ge 4/25$, i.e. $u \le 5/2$, for $t \ge \frac{5}{2}\log(8/3)$, and $\frac{5}{2}\log(8/3) < \frac{5}{2} < 9$ since $8/3 < e$. So all upper bounds hold from $t = 9$ on, and persist.

Stage a. By (16), $a(t) \ge 2/5$ for all $t \ge 9$; also $a \ge 1/10$ for all $t \ge 0$.

Stage b. The clock σ_b has $\sigma_b' = a \ge 1/10$ a.e. on $[0, 9]$ and $\sigma_b' \ge 2/5$ a.e. on $[9, \infty)$, so for $t \ge 9$

$$\sigma_b(t) \;\ge\; \frac{9}{10} + \frac{2}{5}\,(t - 9),$$

and the right side equals $19/5$ at $t = 65/4$. By (18) and (19), $z_b(t) \le 4 + 96e^{-19/5} < 4 + 96 \cdot \frac{3}{128} = \frac{25}{4}$ for $t \ge 65/4$, that is $b(t) > 2/5$ for all $t \ge 65/4$.

Stage c. Now $\sigma_c' = b \ge 1/10$ a.e. on $[0, 65/4]$ and $\sigma_c' \ge 2/5$ a.e. on $[65/4, \infty)$, so for $t \ge 65/4$

$$\sigma_c(t) \;\ge\; \frac{65}{40} + \frac{2}{5}\Big(t - \frac{65}{4}\Big),$$

which equals $19/5$ at $t = 347/16$. The same comparison gives $c(t) > 2/5$ for all $t \ge 347/16$. Finally $347/16 < 22$. □

The two clock identities $\frac{9}{10} + \frac{2}{5}(\frac{65}{4} - 9) = \frac{19}{5}$ and $\frac{65}{40} + \frac{2}{5}(\frac{347}{16} - \frac{65}{4}) = \frac{19}{5}$ are exact. Keeping the exact stage times instead of rounding gives $T_a = 5\log 6$, $T_b = \frac{3}{4}T_a + \frac{5}{2}\log\frac{128}{3}$ and $T_c = \frac{3}{4}T_b + \frac{5}{2}\log\frac{128}{3}$, approximately 8.959, 16.103 and 21.461; neither this nor $347/16$ is claimed optimal. Table 1 summarizes the deadlines.

Requirement	Deadline (exact)	Decimal	Rounded, certified
$a \in [2/5, 5/2]$	$5\log 6$	8.95880	9
$b, c \le 5/2$	$\frac{5}{2}\log(8/3)$	2.45207	9
$b, c \ge 2/5$, one stage	$10\log(128/3)$	37.53418	38
$b \ge 2/5$, staged	$\frac{3}{4}\,5\log 6 + \frac{5}{2}\log(128/3)$	16.10264	$65/4 = 16.25$
$c \ge 2/5$, staged	$\frac{3}{4}T_b + \frac{5}{2}\log(128/3)$	21.46053	$347/16 = 21.6875$

Table 1: Sufficient recovery deadlines in nondimensional time, uniform over all measurable histories in $[1/2, 2]^6$ and all startups in $[1/10, 10]^3$. The decimal column only evaluates the exact formulas; the certified column uses the rational roundings proved in Lean.

What is machine-checked here. The steps of the proof of Theorem 7.3 after the clocks and the inverse-square transformation are formalized as one Lean theorem, `staged_coordinate_recovery` (Section 8): given an absolutely continuous coordinate u with values in $[1/10, 10]$, an absolutely continuous clock σ with $\sigma(0) = 0$ and $\sigma' \ge 0$, coefficients with $k_+ \ge 1/2$, $k_- \le 2$, the equation $u' = \sigma' u(k_+ - k_- u^2)$ almost everywhere, clock floors $\sigma' \ge 1/10$ on $[0, t_1]$ and $\sigma' \ge 2/5$ on $[t_1, \infty)$, and the arithmetic condition $t_1/10 + \frac{2}{5}(t_2 - t_1) \ge 19/5$, it concludes $u(t) \ge 2/5$ for all $t \ge t_2$. Its instances `staged_recovery_second` ($t_1 = 9$, $t_2 = 65/4$) and `staged_recovery_third` ($t_1 = 65/4$, $t_2 = 347/16$) are the two stages, and `a_stage_deadline_lt_nine`, `recovery_deadline_lt_38` and `staged_deadline_lt_22` are the roundings. The conventional parts are the invariance of the startup box, the existence of the clocks with the stated almost-everywhere derivatives (the integral of a bounded measurable function), the identification $\sigma_b' = a$ with the floors from stage a, and the Riccati comparison (15). Figure 2 shows illustrative trajectories under three declared rate schedules; every sampled state after $347/16$ lies in the operating box, as Theorem 7.3 requires, and the pictures show how far from tight the deadline is for these particular histories.

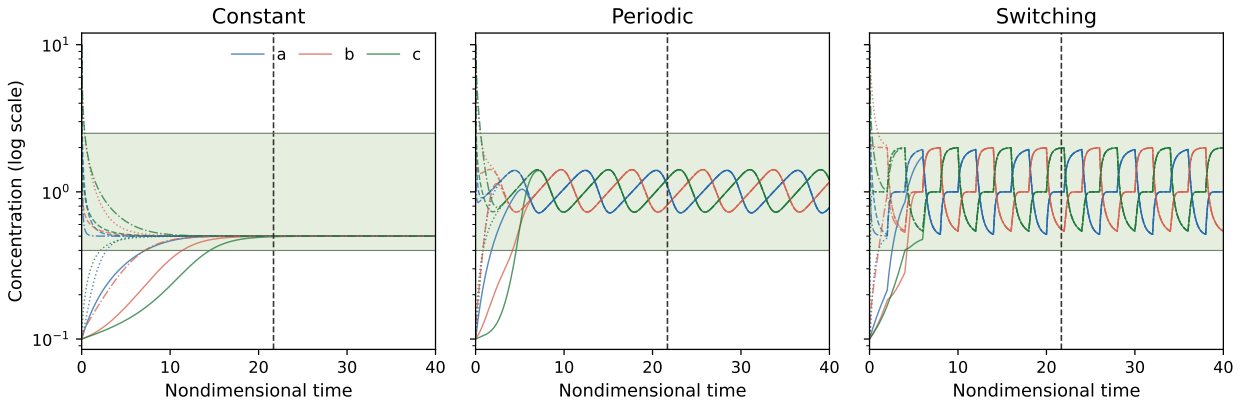


Figure 2: Illustrative trajectories under three declared schedules. The shaded strip is the proved operating interval $[0.4, 2.5]$ for each coordinate and the dashed line is the staged deadline $347/16$. Colours distinguish a, b, c; line styles distinguish the four startups $(0.1, 0.1, 0.1)$ (solid), $(10, 10, 10)$ (dashed), $(0.1, 10, 0.1)$ (dotted) and $(10, 0.1, 10)$ (dash-dotted). Constant rates are $(0.5, 2, 0.5, 2, 0.5, 2)$; periodic rates are $k_i(t) = 1.25 + 0.75\sin(2\pi t/8 + (i-1)\pi/3)$; switching rates are 0.5 when $(\lfloor t/2 \rfloor + i - 1) \bmod 3 = 0$ and 2 otherwise, integrated segment by segment at the exact switching times (SciPy LSODA, relative tolerance 10^{-9}, absolute tolerance 10^{-11}). These simulations illustrate the theorem and supply no all-history evidence.

7.5 Dimensions and physical reading

Let $A = X_* a$, $B = X_* b$, $C = X_* c$ and $t_{\text{phys}} = t_* t$ for declared positive reference scales. The corresponding physical effective model is

$$\dot{A} = C(q_1 - q_2 A^2), \qquad \dot{B} = AB(q_3 - q_4 B^2), \qquad \dot{C} = BC(q_5 - q_6 C^2),$$

with $k_1 = t_* q_1$, $k_2 = t_* X_*^2 q_2$, $(k_3, k_5) = t_* X_* (q_3, q_5)$ and $(k_4, k_6) = t_* X_*^3 (q_4, q_6)$. The q_i carry different units (q_1: time^{-1}; q_2: $\text{concentration}^{-2}\text{time}^{-1}$; q_3, q_5: $\text{concentration}^{-1}\text{time}^{-1}$; q_4, q_6: $\text{concentration}^{-3}\text{time}^{-1}$), so the common dimensionless box $[1/2, 2]^6$ is an assumption on scaled coefficients, not a statement that unlike physical constants share a numerical range; whether useful X_*, t_* exist for given measured ranges must be checked case by case. The physical startup and operating boxes are $[0.1X_*, 10X_*]^3$ and $[0.4X_*, 2.5X_*]^3$, and the deadline is $22t_*$; no calibration of t_* is inferred.

The reduced full-rank system is an open, driven effective model. One mass-balanced bookkeeping realization replaces the forward reactions by $F_A + C \to A + C$, $F_B + A + B \to A + 2B$, $F_C + B + C \to B + 2C$ and the backward ones by $2A + C \to A + C + W_A$, $A + 3B \to A + 2B + W_B$, $B + 3C \to B + 2C + W_C$, with maintained feed reservoirs $F_\bullet$ and removed waste $W_\bullet$ absorbed into the effective coefficients. Such bookkeeping explains how bounded positive effective coefficients can be maintained; it does not establish elementary molecular mechanisms, detailed balance or thermodynamic consistency, and the cubic terms should be read as effective cooperative kinetics. Retention of all species in a positive box implies neither net useful production nor the absence of futile cycles, and it is a deterministic guarantee, not a statement about finite-copy stochastic extinction.

8 Formal verification and reproducibility

The scientific development uses Lean 4 (`leanprover/lean4:v4.30.0`) and Mathlib commit `c5ea00351c28e24afc9f0f84379aa41082b1188f`. Historical source bytes and receipts are preserved. Fresh additive checks on October 8, 2026 reuse exactly matching historical dependency artifacts; they compile the new modules and check the new roots. Warnings are errors and every selected root reports only `propext`, `Classical.choice` and `Quot.sound`. This scientific verification is distinct from migration and mechanical verification of a Palomar submission.

The affine construction (Lemma 3.1) is ported from the public Lean sources of [14, 15], at the cited commit, into `proofs/EndotacticPermanence/Imported/`. The relevant declarations `OAI.Problem326.Affine.exists_fullCertificate` and `FullCertificate.uniform_activity` have no weak-reversibility hypothesis. The proofs and upstream hashes are included; no fixed-point or geometric axiom was added.

The fresh roots. The signed model has a finite reaction index R with source and target maps $R \to \texttt{Fin d} \to \mathbb{R}$. Its `Endotactic` is Definition 2.1, testing every real direction. `PositiveRateBox` has positive lower bounds and coordinatewise upper bounds. `AEAdmissible` means coordinatewise a.e. measurability and a.e. box membership for Lebesgue measure restricted to $[0, \infty)$. `IsCaratheodorySolution` is exactly initial data, absolute continuity on every $[0, T]$, and the vector ODE a.e. on $(0, T)$.

In the namespace `EndotacticPermanence.RealComplex`, `endotactic_permanence_compact_ae` (`Real/AlmostEverywherePermanence.lean`) proves Theorem 2.2 in this signed domain. Its quantifier order is $\exists K\ \forall Q\ \exists T_Q\ \forall x_0 \in Q\ \forall k$. It concludes global positive existence, class preservation, absorption, uniqueness among positive solutions, and invariance for every positive solution starting in K. The pointwise-rate root is `RealComplex.endotactic_permanence_compact`. Positive-box regularity and clamped-field existence replace ambient polynomial regularity; uniqueness uses finite-time positive box enclosures.

`RealComplex.mass_action_continuous_dependence_ae` (`Real/ContinuousDependence.lean`) proves Proposition 6.1 with its Lipschitz constant chosen before the history and curves. `EndotacticPermanence.repair_rate_history` constructs a common null-set repair, and `caratheodory_solution_congr_forward_ae` transfers the solution predicate for every curve. The fresh natural-complex root `EndotacticPermanence.endotactic_permanence_compact_ae` additionally preserves the historical theorem's ambient uniqueness conclusion.

Example correspondence. The historical `RecoveryComparison.lean` checks clocked comparisons, the inverse-square transformation, two-stage clock floors and coordinate recovery. Its principal composed declaration is `staged_coordinate_recovery`, with applications `staged_recovery_second` and `staged_recovery_third`. The exact roundings are `a_stage_deadline_lt_nine`, `recovery_deadline_lt_38` and `staged_deadline_lt_22`. These remain conditional on the explicitly listed curve and clock hypotheses in Section 7.3.

Remaining formal boundaries. Periodic and equilibrium corollaries require a finite-dimensional fixed-point theorem and its solution-map interfaces. Feedback needs a local closed-loop existence and continuation proof. Those results, and the fixture-specific invariance and Riccati comparisons, are conventional here. They are not part of the compared formal claim. No custom axiom was introduced to close them.

Result	Verification status
Signed real-complex permanence, a.e. admissible histories: global positive solutions, positive-solution uniqueness, invariant compact convex K, compact-initial uniform deadline	Fresh strict Lean root
Natural-complex permanence with a.e. rates and ambient uniqueness	Fresh strict Lean root
Initial-data continuous dependence on a compact positive region, signed complexes and a.e. rates	Fresh strict Lean root
Natural-complex pointwise-rate roots and continuous dependence	Preserved historical checks
Feedback, periodic operation and autonomous equilibria	Conventional proofs
Example: classification, invariant boxes, clock construction, Riccati comparison	Conventional proofs
Example: clocked comparisons, inverse-square transformation, two-stage clock floors and exact roundings	Preserved historical checks

Table 2: Formal and conventional scope. No conditional fixed-point statement is counted as a proved corollary.

Reproduction. The accompanying repository includes the manuscript sources, bibliography, figures, figure producer and saved trajectory data, together with the Lean development migrated to Lean 4.35.0-rc2. The independent statement in `Challenge.lean` and its implementation in `Solution.lean` select uniform permanence and initial-data continuous dependence. Both declarations passed local Comparator and all six bundled kernel checks. The report `verification/local-check.json` records the exact source hashes, checked declarations and an extract of the saved results. These checks are separate from the scientific Lean 4.30.0 checks described above. Build instructions are in `BUILD.md`; pinned upstream attribution is in `UPSTREAM_PROVENANCE.json`. The additional corollaries retain the formal and conventional scope stated in Table 2.

9 Limits and open questions

The theorem permits arbitrary measurable switching inside a fixed positive compact box, and its constants depend on that box; neither bound can simply be dropped. For the endotactic pair $A \rightleftarrows 2A$ with common history $k_+(t) = k_-(t) = e^{-t}$, the operational clock has finite total length and the logistic solution satisfies $a(t) \le e\, a_0$, so arbitrarily small startups never reach a common positive floor. For $0 \rightleftarrows A$ with birth rate $1+t$ and death rate 1, $a' = 1+t-a$ grows without bound. These examples delimit the hypothesis; they do not weaken the classwise statement under it.

The construction supplies a finite sum of barrier delays, not an efficient algorithm; making the affine data effective, and extracting network-specific recovery constants of the kind obtained in Section 7, are quantitative problems. The staged argument there illustrates a general principle that may deserve a theorem of its own: a recovered component accelerates the guaranteed recovery of the components it drives, without any reduction of the rate uncertainty, provided the dependency order is respected. Positive deterministic permanence says nothing by itself about extinction in finite stochastic populations; periodic-solution existence leaves stability, entrainment and uniqueness open; and the feedback proposition requires its local hypotheses. Which quantitative conclusions survive as a lower rate bound tends to zero, and under what monotonicity conditions periodic existence becomes entrainment, are natural next questions.

A Lean statement of the signed root

The fresh compact-initial root is reproduced below, transliterated from Unicode to ASCII. Its namespace is `EndotacticPermanence.RealComplex`; `AEAdmissible` is defined in Section 8. The source file and the elaborated declaration in its receipt are authoritative. The preserved natural-complex statements remain included in the source bundle.

```
theorem endotactic_permanence_compact_ae {d : Nat} {R : Type*} [Fintype R]
    (source target : R -> Fin d -> Real)
    (endo : Endotactic source target) (B : PositiveRateBox R)
    (c : Fin d -> Real) (hc : PositiveState c) :
    exists K : Set (Fin d -> Real), K.Nonempty /\ IsCompact K /\ Convex Real K /\
      K subset positiveClass source target c /\
      (forall Q : Set (Fin d -> Real), IsCompact Q -> Q subset positiveClass source
        target c ->
        exists T : Real, 0 <= T /\ forall x0 in Q, forall k : Real -> R -> Real,
          AEAdmissible B k -> exists x : Real -> (Fin d -> Real),
            IsCaratheodorySolution (fun t => indexedMassAction source target (k t)) x0 x
              /\
            (forall t : Real, 0 <= t -> x t in positiveClass source target c) /\
            (forall t : Real, T <= t -> x t in K) /\
            (forall y : Real -> (Fin d -> Real),
              IsCaratheodorySolution (fun t => indexedMassAction source target (k t)) x0
                y -> (forall t : Real, 0 <= t -> PositiveState (y t)) ->
                EqOn x y (Ici 0))) /\
      (forall x0 in K, forall k : Real -> R -> Real, AEAdmissible B k ->
        forall x : Real -> (Fin d -> Real),
          IsCaratheodorySolution (fun t => indexedMassAction source target (k t)) x0 x
            -> (forall t : Real, 0 <= t -> PositiveState (x t)) ->
          forall t : Real, 0 <= t -> x t in K)
```

References

[1] David F. Anderson. A proof of the Global Attractor Conjecture in the single linkage class case. *SIAM Journal on Applied Mathematics*, 71(4):1487–1508, 2011.

[2] David Angeli, Patrick De Leenheer, and Eduardo D. Sontag. A Petri net approach to the study of persistence in chemical reaction networks. *Mathematical Biosciences*, 210(2):598–618, 2007.

[3] Balázs Boros and Josef Hofbauer. Permanence of weakly reversible mass-action systems with a single linkage class. *SIAM Journal on Applied Dynamical Systems*, 19(1):352–365, 2020.

[4] Earl A. Coddington and Norman Levinson. *Theory of Ordinary Differential Equations*. McGraw-Hill, New York, 1955.

[5] Gheorghe Craciun. Toric differential inclusions and a proof of the global attractor conjecture. arXiv:1501.02860v3, 2015. Version 3, revised September 23, 2026.

[6] Gheorghe Craciun and Abhishek Deshpande. Endotactic networks and toric differential inclusions. *SIAM Journal on Applied Dynamical Systems*, 19(3):1798–1822, 2020. arXiv:1906.08384.

[7] Gheorghe Craciun, Fedor Nazarov, and Casian Pantea. Persistence and permanence of mass-action and power-law dynamical systems. *SIAM Journal on Applied Mathematics*, 73(1):305–329, 2013. arXiv:1010.3050.

[8] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635, 2021.

[9] Martin Feinberg. Complex balancing in general kinetic systems. *Archive for Rational Mechanics and Analysis*, 49:187–194, 1972.

[10] A. F. Filippov. *Differential Equations with Discontinuous Righthand Sides*. Kluwer, Dordrecht, 1988.

[11] Manoj Gopalkrishnan, Ezra Miller, and Anne Shiu. A geometric approach to the Global Attractor Conjecture. *SIAM Journal on Applied Dynamical Systems*, 13(2):758–797, 2014. arXiv:1305.5303.

[12] F. Horn and R. Jackson. General mass action kinetics. *Archive for Rational Mechanics and Analysis*, 47:81–116, 1972.

[13] Matthew D. Johnston, Casian Pantea, and Pete Donnell. A computational approach to persistence, permanence, and endotacticity of biochemical reaction systems. *Journal of Mathematical Biology*, 72(1–2):467–498, 2016. arXiv:1412.4662.

[14] OpenAI. Boundedness and persistence of weakly reversible mass-action systems. OpenAI Math Release preprint, September 25, 2026. Paper and Lean sources in the public repository `https://github.com/openai/math` at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`: `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Boundedness-and-persistence-of-weakly-reversible-mass-action-systems-September-25-2026` and `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/lean/OAI/Analysis/MassAction`, 2026.

[15] OpenAI. Uniform permanence in weakly reversible mass-action systems. OpenAI Math Release preprint, October 5, 2026. Public repository `https://github.com/openai/math` at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`: `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Uniform-Permanence-in-Weakly-Reversible-Mass-Action-Systems-October-5-2026`, 2026.

[16] Casian Pantea. On the persistence and global stability of mass-action systems. *SIAM Journal on Mathematical Analysis*, 44(3):1636–1673, 2012. arXiv:1103.0603.

[17] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381, 2020.